LAB 11

Partial Differential Equation

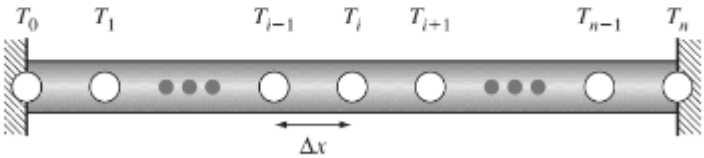
Assignment:

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Consider 1D heat conduction in metal rod as below.



1D heat conduction governing equation is given below,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2}\right)$$

Write down the discretized equation is explicit finite difference method and obtain the Temperature profile T(x,t). Take length of rod is 0.75 m, Δx= 0.025 m, $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

Hand Calculation

$$\frac{\partial T}{\partial t} = \frac{k\,\partial^2 T}{\partial x^2}$$

$$\frac{\partial T}{\partial t} = \frac{T^{n+1} - T^{n}}{\Delta t} \quad \text{—(i)}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1}^{n} - 2T_{i}^{n} + T_{i-1}^{n}}{\Delta x^2} \quad \text{—(ii)}$$

Substituting ① & ② in the initial equation

$$T_i^{n+1} = \left[T_i^{n} + \frac{k\Delta t}{\Delta x^2}\left(T_{i+1}^{n} + 2T_i^{n} + T_{i-1}^{n}\right)\right]$$

Figure 1 hand calculation of heat conduction

Matlab Code

/MATLAB Drive/untitled15.m

```
% Parameters
L = 0.75; dx = 0.025; dt = 0.00015;
K = 1; eps = 1e-4;
N = L/dx + 1;

% Initialization
T = zeros(1,N);
T(1) = 100;   % left end
T(N) = 0;     % right end

error = 1;

while error > eps
    T_old = T;

    for i = 2:N-1
        T(i) = T_old(i) + K*dt/dx^2 * ...
              (T_old(i+1) - 2*T_old(i) + T_old(i-1));
    end

    error = max(abs(T - T_old));
end

plot(T,'r-o'); grid on;
title('1D Heat Conduction');
xlabel('x'); ylabel('Temperature');
```

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Figure 2 Matlab Code for head conduction

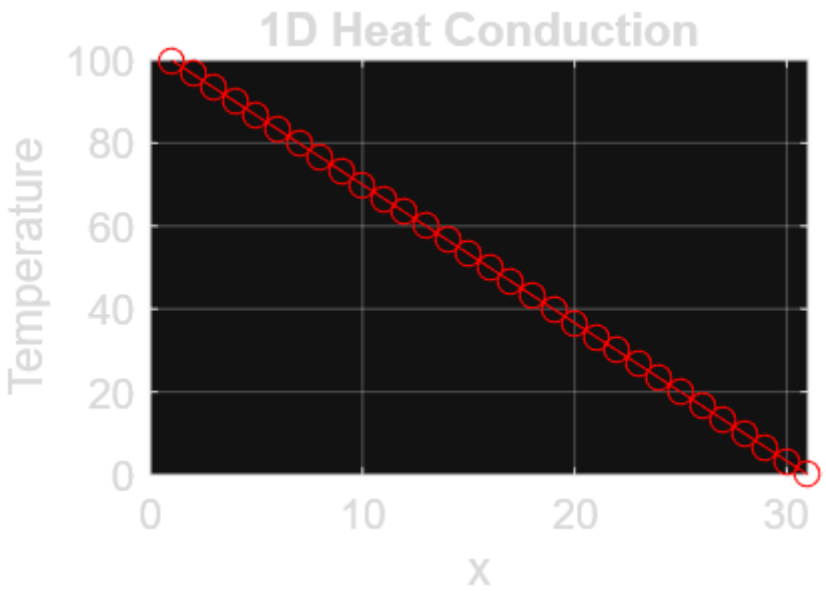

Figure 3 Graph of 1d Heat conduction